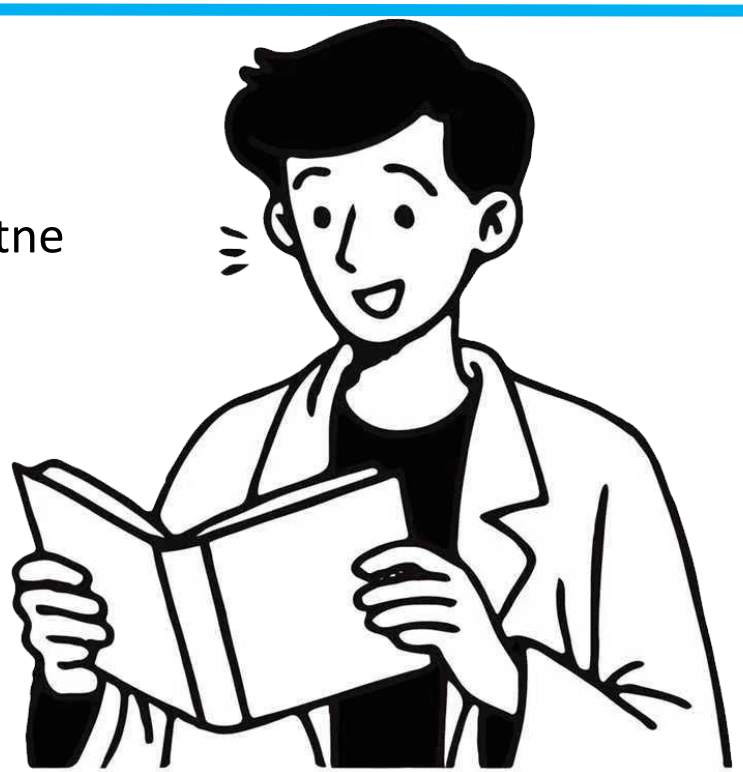
Open-RAN Architecture, Interfaces, Algorithms, Security and Research Challenges

Supervisor: Dr. Chatura Seneviratne
Co-Supervisor 1: Prof. Dr. An Braeken
Co-Supervisor 2: Mr. Pramitha Fernando



EG/2021/4474 Dewmith M.K.J.
EG/2021/4485 Dilshan M.D.K.
EG/2021/4807 Sewinda L.L.D.
EG/2021/4832 Thilohith T.K.

21/11/2025

CONTENT

01 | Introduction to O-RAN

02 | Architecture & Principles

03 | Near-RT RIC, xApps

04 | Non-RT RIC, rApps

05 | Key O-RAN Interfaces

06 | Built-in AI/ML Workflow

07 | Security Analysis

08 | Research Platforms

PART 01

Why Open RAN?

Limitations of Legacy RAN

Monolithic Structure

Traditional RANs are monolithic units (black boxes), that bundles all functionalities from PHY to RRC in a single vendor's hardware. This closed architecture limits flexibility and customization, making it difficult to optimize for diverse deployments and traffic profiles.

Vendor Lock-In

Operators are often locked in to a single vendor's solution, with limited options for interoperability. This restricts competition and innovation, driving up costs and reducing the ability to adopt new technologies quickly.

Operational Challenges

Real-time optimization and efficient spectrum utilization are challenging due to the lack of open interfaces and programmability. Upgrades and maintenance require significant effort and cost, further burdening operators.

PART 02

Architecture & Principles

Four Key Principles of O-RAN

- Disaggregation
- Intelligent, data-driven control with RICs
- Virtualization
- Open Interfaces

Disaggregation

This involves splitting the traditional base stations into different functional units extending the functional disaggregation paradigm proposed by 3GPP(4G) for the NR(5G) Next Generation Node Bases (gNBs).

- The gNB is split into RU, DU and CU components.
- Each handling different layers of the 3GPP protocol stack.
- This logical split allows different functionalities to be deployed at different locations of the network, as well as on different hardware platforms.
- This modular approach allows for independent scaling and optimization of each component.

CU–DU–RU Functional Split

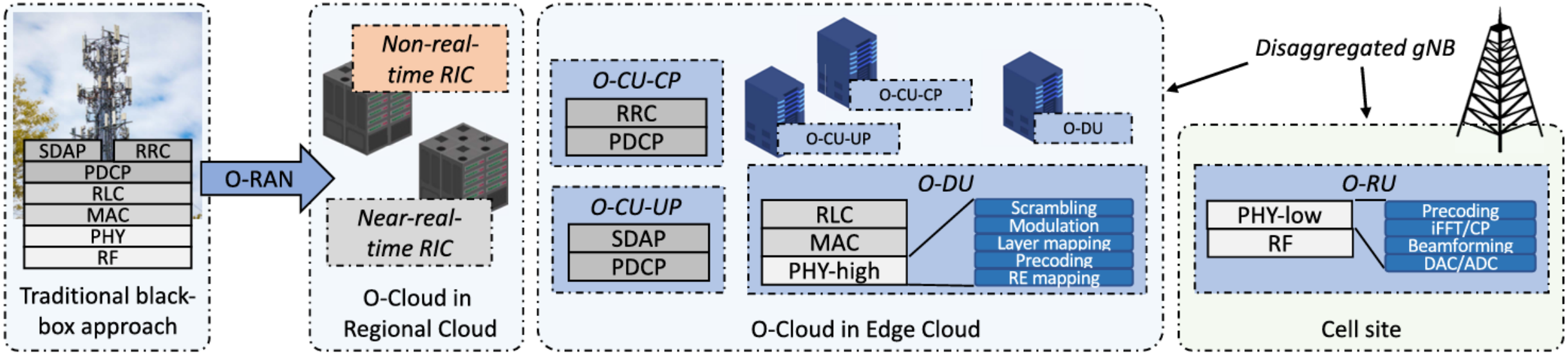


Figure 01: Traditional black-box approach to virtualized gNB with functional split

CU-DU-RU Functional Split

O-RAN Component	Location/Hardware Type	3GPP Protocol Layers	Key Functionalities (7.2x split)
Central Unit (O-CU)	Virtualized (White box server at the edge)	Radio Resource Control (RRC) layer, Service Data Adaptation Protocol (SDAP) layer, Packet Data Convergence Protocol (PDCP) layer	Manages connection lifecycle (RRC), QoS (SDAP), Encryption and reordering (PDCP), Logically split into Control Plane (CP) and User Plane (UP)
Distributed Unit (O-DU)	Virtualized (White box server at the edge)	Radio Link Control (RLC) layer, Medium Access Control (MAC) layer, Higher portion of the Physical layer (High PHY)	Performs frequency domain functionalities including precoding, scrambling, modulation, layer mapping and mapping into physical resource blocks
Radio Unit (O-RU)	Dedicated hardware deployed close to antennas (FPGA, ASICs)	Lower portion of the Physical layer (Low PHY), RF components	Performs time domain functionalities such as FFT, precoding, cyclic prefix addition/removal, beamforming and RF operations. Designed to be inexpensive and easy to deploy

Intelligent, data-driven control with RICs

Introduce programmable components that can run optimization routines with closed-loop control and orchestrate the RAN.

What are RICs?

RICs are central to O-RAN's vision of intelligent, data driven network management. They collect and process vast amounts of telemetry data — called **Key Performance Measurements (KPMs)**

RIC Type	Location	Control Loop Timing	Role
Non-Real-Time RIC	In the SMO (Service Management and Orchestration) framework	> 1 s	Provides strategic guidance, ML model management and policy deployment
Near-Real-Time RIC	At the network edge and operates control loops periodically	10 ms to 1 s	Executes tactical control actions like handovers, load balancing and scheduling

Virtualized

Network functions are implemented as software running on commodity hardware, leveraging cloud-native principles. This enables flexible deployment, scalability and cost savings compared to proprietary hardware solutions.

O-RAN introduces cloud-based components for **network management & optimization** **Near-RT and Non-RT RICs** run on a hybrid cloud platform called **O-Cloud** = **pooled compute + virtualization** across data centers. This combines: physical nodes, OS, hypervisors, orchestration tools. Enables,

- Decoupling between hardware and software components
- Harmonization and definition of the hardware capabilities for the O-RAN infrastructure
- Sharing of the hardware among different tenants
- Automated deployment and instantiation of RAN functionalities.

Open Interfaces

Open RAN promotes standardized public interfaces, allowing different vendors to interoperate seamlessly. This transparency fosters competition and innovation, breaking the traditional vendor lock-in model.

O-RAN defines **standardized open interfaces** connecting all RAN components that enables **multi-vendor interoperability** (RIC ↔ CU/DU/RU) by breaking traditional **vendor lock-in.** This gives operator to **full access & control** over RAN data and functions and supports **flexible, software-driven radio resource management**

Encourages:

✓ Multi-vendor interoperability

✓ Innovation accelerates

✓ Update cycles shorten

✓ Market competition increases

O-RAN, vRAN and Traditional RAN

Traditional RAN

The starting point, defined by its **monolithic, proprietary** nature, resulting in vendor lock-in.

vRAN/C-RAN

Introduced the concept of **virtualization**, centralizing baseband processing to leverage cloud data centers for efficiency and cost reduction.

O-RAN

Created when the C-RAN Alliance and the xRAN Forum joined forces in 2018. Builds directly on vRAN/Cloud RAN principles by adding three critical elements:

- Openness: Defining **open interfaces** to ensure different vendor components are interoperable
- Disaggregation: Formally **splitting the base station into functional units** (CU, DU, RU)
- Intelligence: **RAN Intelligent Controllers (RICs)** to enable **intelligent, data-driven control** over the network

O-Cloud & Containerization

The O-Cloud is the enabling hybrid cloud computing platform for the O-RAN architecture. This consists of:

- Pooled computing resources and virtualization infrastructure across edge and regional data centers, hosting CUs, DUs and RICs as microservices on Kubernetes.
- This enables scalable, cloud-native deployment and efficient resource management, optimizing power consumption by limiting power use only to the network functions actually needed.

Role of O-RAN Alliance vs 3GPP

01 O-RAN Alliance

O-RAN Alliance was created in 2018, with the goal to specify and standardize an intelligent, open virtualized and interoperable RAN architecture. The Alliance has several working groups (WGs) responsible for different specifications such as WG3 for the near-RT RIC and E2 interface, WG4 for the Open Fronthaul Interface.

02 3GPP

The O-RAN Alliance builds upon existing 3GPP standards while 3GPP defines the core radio protocol stack layers and several Intra-RAN interfaces like F1, Xn and E1. O-RAN embraces and extends this architecture by adding programmable components (RICs) and defining necessary open interfaces (E2, A1, O1, O-FH) to connect disaggregated and intelligent components from multiple vendors.

PART 03

Near-RT RIC, xApps

Near RT-RIC

Responsible:

The near-RT RIC is a logical function which responsible for **real-time optimization** of RAN operations.

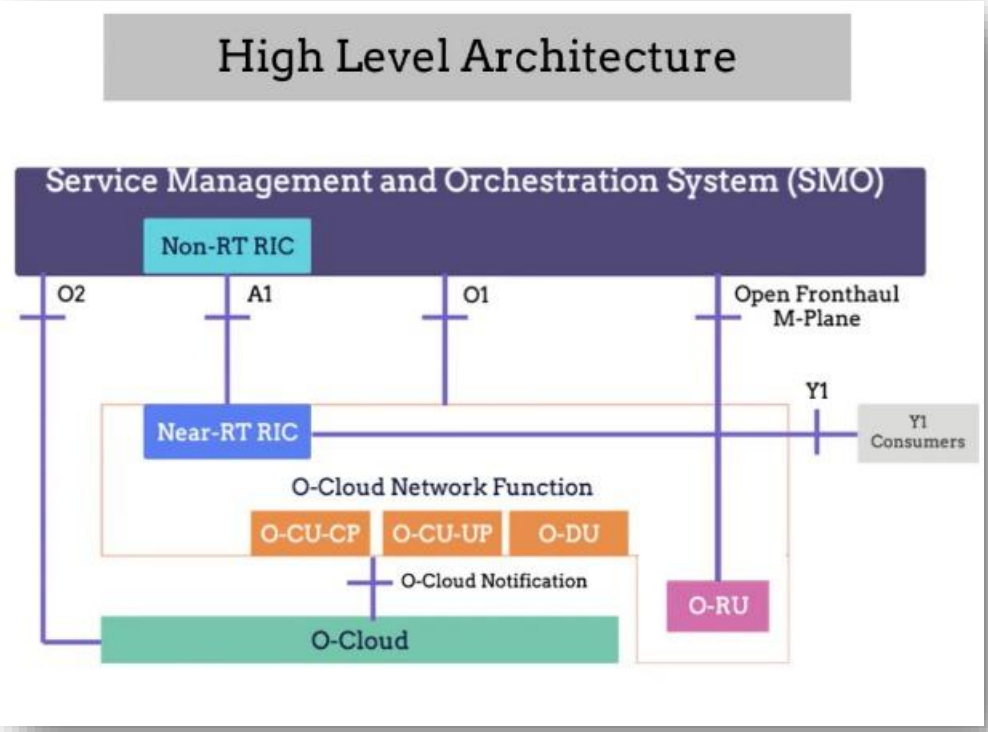
Operates:

It operates on time scales between **10 milliseconds and 1 second**, allowing it to respond quickly to changing network conditions.

Interacts:

This interacts directly with RAN nodes such as:

- **Central Units (CUs)**
- **Distributed Units (DUs)**
- **Legacy LTE eNBs (4G Base stations)**



Enables near-RT RIC to influence key performance aspects like handovers, load balancing and resource scheduling.

Ex: Consumers can experience QoS, UE location…etc. From Near-RT RIC

Near RT-RIC Internal Components

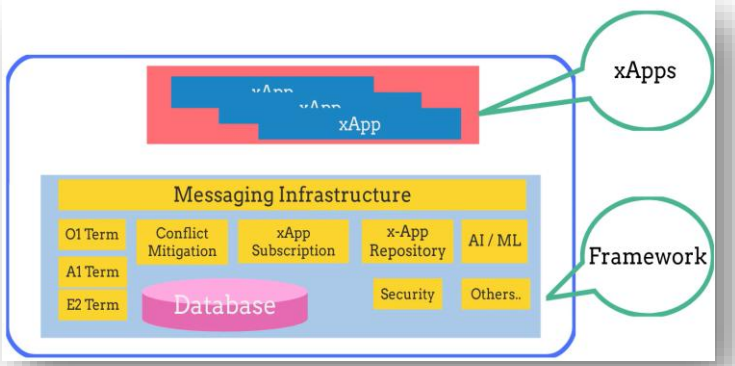
Component	Purpose & Functionality
Internal Messaging Infrastructure	Connects xApps, platform services, and interface terminations to each other. Supports registration, discovery, deletion of endpoints, and message routing via point-to-point or publish/subscribe.
Conflict Mitigation	Detects and resolves conflicts among xApps. Handles direct (e.g., conflicting settings), indirect (e.g., resource contention), and implicit (e.g., performance trade-offs) conflicts.
Subscription Manager	Manages xApp subscriptions to E2-exposed functions. Controls access to E2 messages and merges duplicate subscription requests to reduce overhead.
Security Sub-System	Aims to prevent malicious xApps from leaking sensitive data and degrading RAN performance.
Network Information Base (NIB) & Shared Data Layer (SDL)	R-NIB database stores E2 nodes.(The UE identity is a key and sensitive piece of information in the RIC,) tracks and controls UE identities across nodes. Database can be queried by the different components of the RIC platform (including the xApps) through the Shared Data Layer (SDL) APIs.
xApp Management	Automates xApp lifecycle: onboarding, deployment, termination (via SMO). Supports FCAPS (Fault, Configuration, Accounting, Performance, Security) logging

Near RT-RIC – xApps Role

xApps Role:

An xApp is a plug and play microservice that can be used to perform radio resource management. (Optimization algorithms)

Ex: In RAN data analysis and RAN control, xApps can receive data and telemetry from the RAN and send back control using the E2 interface.



*** Telecom Infra Project (TIP) RAN Intelligence & Automation (RIA)** subgroup pushing for expanded APIs and interoperability

To supports xApps, Near-RT-RIC include Framework,

(i) A database containing information on the RAN (e.g., list of connected RAN nodes, users, etc.) and serving as a shared data layer among xApps.

(ii) Messaging infrastructure across the different components of the platform, also supporting the subscription of RAN elements to xApps.

(iii) Terminations for open interfaces and Application Programming Interfaces (APIs).

(iv) Conflict resolution mechanisms to orchestrate control of the same RAN function by multiple xApps.

How xApps works?

Large amount of data, Counters, Failure Information, Events

Collected from gNBs and used

AI/ML Training

Model to

Learn or abstract

Algorithms - xApps

AI/ML Models

- Networks spatial-temporal traffic patterns.
- User mobility patterns.
- Service type patterns.
- Network Slice patterns.
- Network QoS prediction patterns.
- MIMO parameters configurations.

Other Ran logic algorithms

+

Network wide Context

Policies can be

framed to drive

Network grain operations

- Radio Resource management
- Load balancing
- ID allocations
- Handover decisions

Open Source Near-RT RIC Implementations

An xApp consists of 2 key components,

1. xApp Software Image

The complete set of files required to deploy and run the xApp. (**Docker image** deployed on a **Kubernetes** cluster)

2. xApp Descriptor (YAML/JSON)

Defines how the xApp should be managed. Includes configurations such as:

- Autoscaling policies
- Deployment and deletion procedures
- Upgrade information

Implementations

- **OSC RIC** - Official reference implementation, baseline for experimentation.
- **ColO-RAN** - Research-focused, built on OSC RIC.
- **SD-RAN (ONF)** - Cloud-native, ONOS-based, supports standard + custom E2SMs, includes SDK and xApps.
- **FlexRIC** - Monolithic RIC, interfaces with Open Air Interface, offers monitoring/slicing service models + SDK.
- **5G-EmPOWER** - Heterogeneous RANs, adds Wi-Fi + cellular mobility, multi-tenant support, custom slice resource allocation.

PART 04

Non-RT RIC, rApps

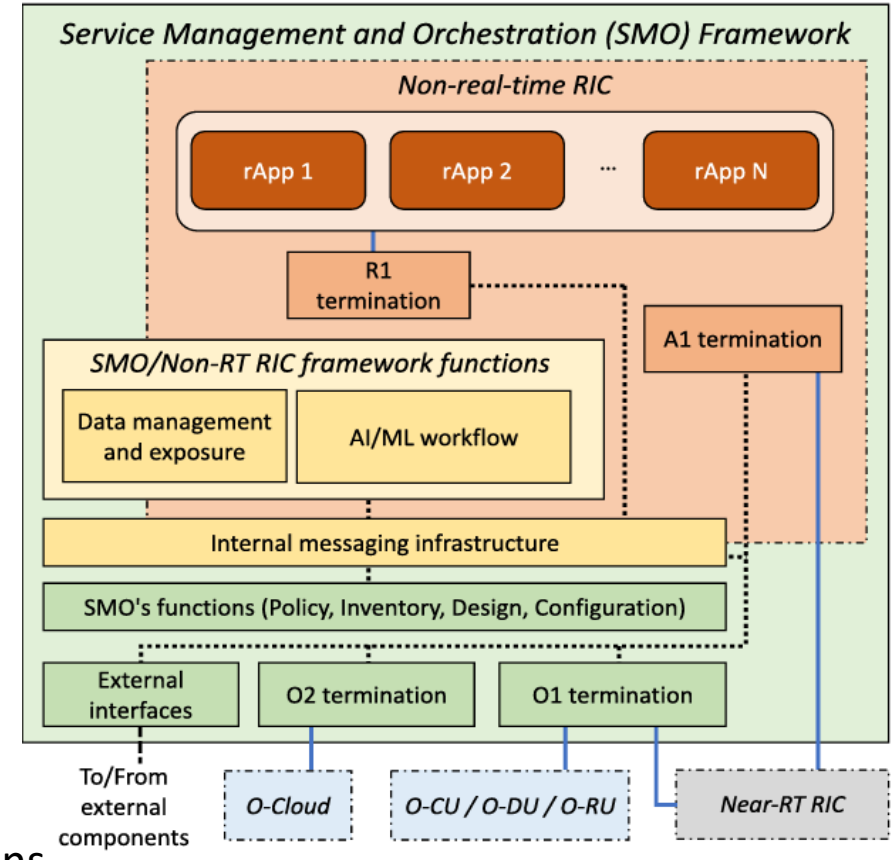
Non-RT-RIC

Responsible:

- This component is in charge of handling all orchestration, management and automation procedures to monitor and control RAN components at timescales **> 1 second.**

Supports:

- Interaction between the different network components as well as data collection capabilities to facilitate network monitoring and control via AI/ML



Architecture

O-RAN specs do not strictly define the split between SMO and non-RT RIC functions.

Functions grouped into three sets:

- Orange blocks → Identifies functionalities and interfaces that are anchored to the non-RT RIC.
- Green blocks → Identifies functionalities anchored outside the non-RT RIC.
- Yellow blocks → Span multiple components or not yet anchored.

Non-RT-RIC

Provides two orchestration services:

- **Intent-based network management** → Allow operators to specify intent high-level language through a human-machine interface. YAML/XML configuration files are parsed which determine policies and deployments.
- **Intelligent orchestration** → Coordinates xApps/rApps across RICs, ensures correct location, fed relevant data, conflict-free execution.

Other SMO/Non-RT RIC Functionalities

- **Internal messaging infrastructure** → Links SMO + non-RT RIC, enabling policy/data exchange.
- **Data management & exposure services** → Producer/consumer model for performance reports, AI predictions, etc.
- **AI/ML workflow** → Full lifecycle: data collection, training, validation, deployment, execution.

Non-RT RIC & rApps

1 Supports

rApps can support the same control functionalities provided by xApps. Ex: Traffic steering, Scheduling control, Handover management (at larger timescales.)

2 Hosts

R1 interface → connects rApps to data management, AI/ML, and A1/O1/O2 interfaces.

3 Operates

rApps designed to derive control policies that operate at a higher level and affect a larger number of users and network nodes.

4 rApps use cases

For non-RT RAN control applications include Frequency and interference management, RAN sharing, performance diagnostics, end-to-end Service Level Agreement (SLA) assurance and network slicing.

SMO Framework and Open-Source Implementations

What the SMO Does in Non-RT RIC:

- Handles **FCAPS** (Fault, Configuration, Accounting, Performance, Security)
- Maintains: Service & resource inventory, Topology & network configuration, Policy management for orchestration.

Open-Source Implementations

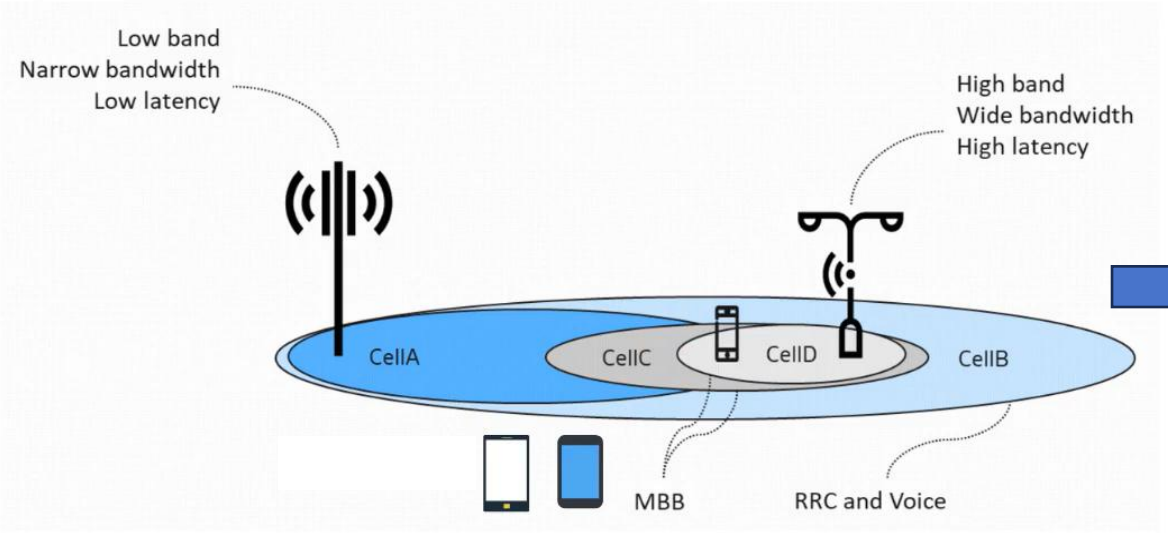
1. ONAP- (*Open Network Automation Platform)*

- Hosted by **Linux Foundation**
- Integrates with Kubernetes, Akraino, Acumos, OpenDaylight.
- Feature-rich but complex
- Used in OSC to build an open-source SMO (monitoring, messaging, event handling)

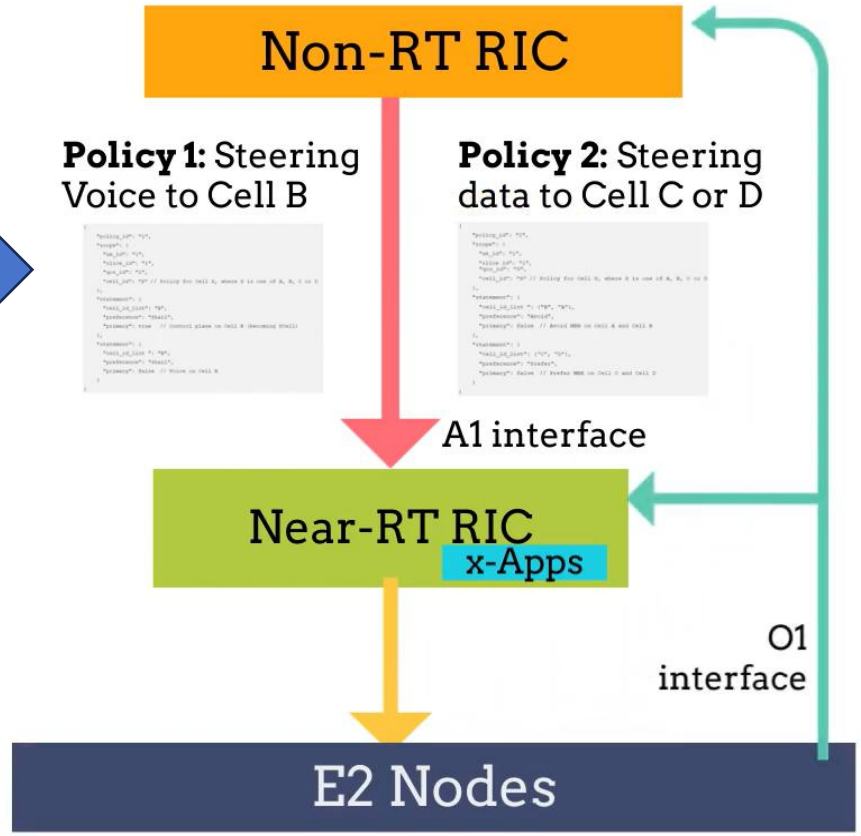
2. OSM- *(Open-Source Management and Orchestration)*

- Hosted by **ETSI** *(European Telecommunication Standards Institute)*
- Based on ETSI **VNF** standards.
- More lightweight than ONAP
- Integration with O-RAN boosted by the **ETSI**

Traffic Steering Example



voice data



Cell A, B

- Lower Frequency Bands
- Good for Voice / less latency

Cell C, D

- HIgh Frequency Bands
- Good for Data / High Bandwidth

PART 05

Key O-RAN Interfaces

Key Interfaces in O-RAN

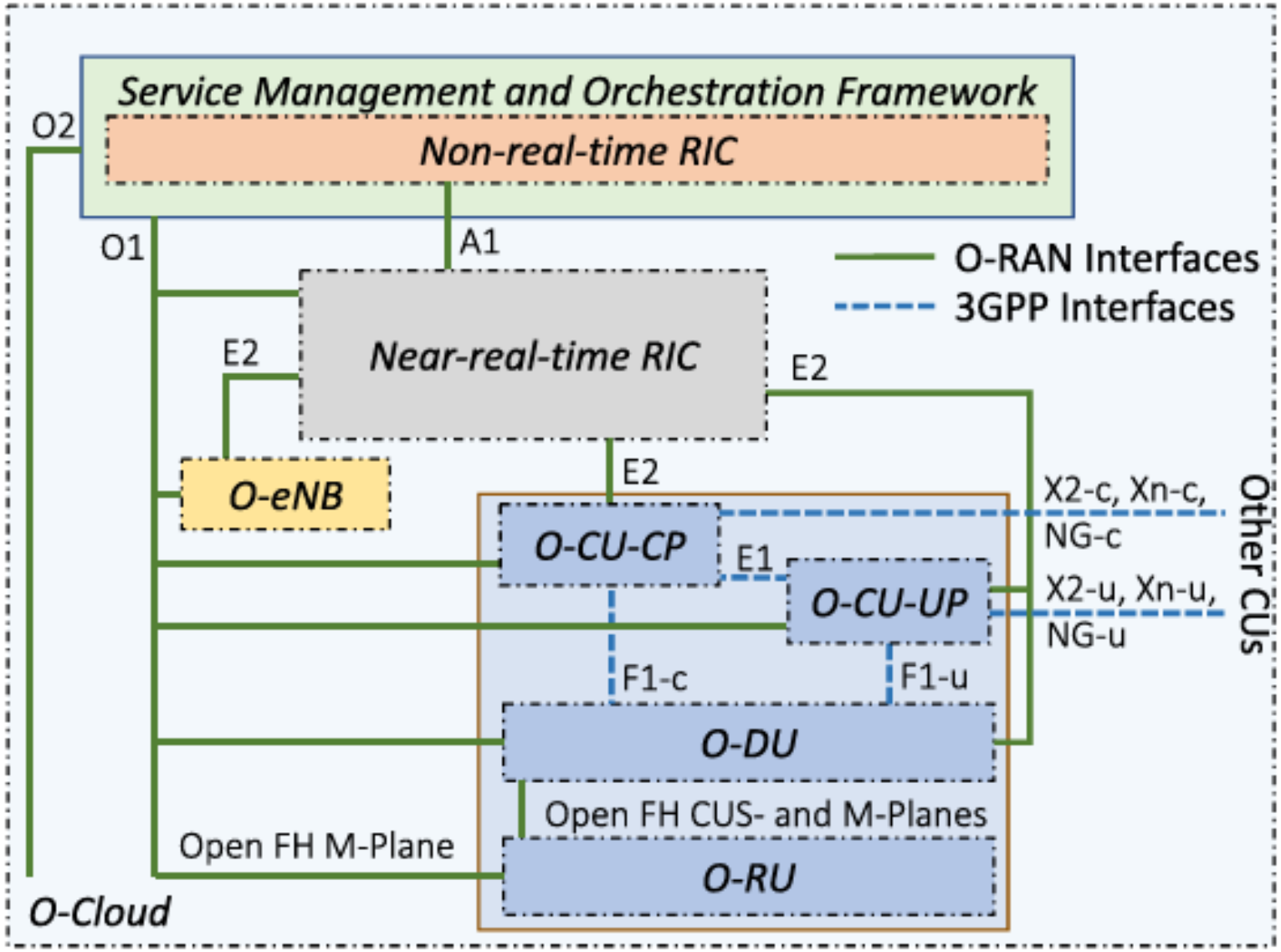


Figure 02: O-RAN architecture, interfaces from O-RAN and 3GPP

Key Interfaces in O-RAN

Interface	Connects	Purpose
E2	Near-RT RIC ↔ RAN nodes (CU, DU, eNBs)	Enables near-real-time control loop by streaming telemetry from the RAN and deploying control actions via specific E2 service models
A1	Non-RT RIC/SMO ↔ Near-RT RIC	Enables non-real-time control loop and the deployment of policy, guidance, manage ML models and transfer enrichment information (EI) to the near-RT RIC
O1	Non-RT RIC/SMO ↔ O-RAN managed elements (RICs, CU, DU, RU)	Supports management services (MnS) including provisioning configuration, life-cycle management and assurance for data analytics and training
O2	SMO ↔ O-Cloud	Management and provisioning of network functions within the cloud platform
Open Fronthaul	DU ↔ RU	Enables 7.2x functional split by transporting data across four planes: User (U-), Control (C-), Synchronization (S-) and Management (M-)

E2 Interface

E2 Functionality

The E2 interface connects the near-RT RIC to RAN nodes (CU/DU/eNB) over SCTP, enabling real-time control and telemetry. It supports setup, report, insert, control, and policy services.

Service Models

E2 service models like KPM, NI, CCC, and RC expose various control capabilities, from performance metrics to UE-level actions, enabling closed-loop optimization with 10 ms–1 s latency.

Real-Time Control

xApps running on the near-RT RIC use E2 to subscribe to KPMs, run ML inference, and send control actions to RAN nodes, enabling dynamic resource management and optimization.

A1 & O1 Interfaces

O1 Interface

The O1 interface connects the SMO to managed elements for FCAPS management, KPI streaming, software upgrades, and trace collection. It uses NETCONF/REST for efficient operations.

O2 Interface

The O2 interface connects the SMO to the O-RAN O-Cloud, enabling management and provisioning of virtualized network functions—including inventory, monitoring, fault tolerance, updates—and is built using NFV standards and cloud platforms like Kubernetes, OpenStack, and ONAP/OSM.

Open Fronthaul Interface

1 › 2 › 3 › 4

Fronthaul Overview

The open fronthaul 7.2x interface connects DUs and RUs over Ethernet, enabling multi-vendor interoperability. It balances complexity, latency, and bandwidth requirements.

Control Plane

The C-plane carries scheduling, beamforming, and control commands from the DU to the RU, ensuring efficient resource management and coordination.

User Plane

The U-plane transports IQ samples between the DU and RU, supporting high-throughput data transfer with compression techniques to optimize bandwidth usage.

Management Plane

The M-plane uses NETCONF for RU configuration and management, enabling secure and efficient deployment and operation of RUs in the network.

PART 06

Built-in AI/ML Workflow

Data Collection & Prep

Open RAN collects data from O1, E2, and A1 interfaces, preparing it for ML training. This includes normalization, augmentation, and dimensionality reduction to ensure high-quality datasets.

Data Sources

Data is collected from O1 KPIs, E2 KPMs, A1 enrichment, and external sources like weather and maps. This diverse data enables comprehensive network insights for ML training.

Data Lakes

Data lakes store historical datasets, providing a rich source for ML model training. Preprocessing ensures data is normalized and ready for use, enhancing model robustness.

Data Collection Sources

Training & Validation

ML models are trained offline using collected datasets. Validation ensures models are reliable and effective before deployment, preventing issues like overfitting or poor performance.

Deployment & Inference

Trained models are deployed as xApps or rApps. Inference runs in real-time on the near-RT RIC, enabling dynamic control and optimization with continuous monitoring for performance.

Deployment & Inference

Model Containerization

Trained ML models are containerized into xApps or loaded as files, ensuring portability and ease of deployment across different RIC implementations.

Performance Telemetry

Performance telemetry is collected continuously, enabling detection of model drift and triggering periodic retraining to maintain accuracy.

Inference Execution

Inference runs on GPUs/CPUs inside the near-RT RIC, meeting strict latency requirements. This enables real-time control and optimization of network resources.

Continuous Learning

The continuous learning loop ensures models adapt to changing network conditions, maintaining optimal performance and preventing degradation over time.

PART 07

Security Analysis

Expanded Threat Surface

New Interfaces

The introduction of open interfaces (Fronthaul, E2, A1, O1, O2) significantly widens the threat landscape, as each interface introduces new, specific vulnerabilities.

xApps and rApps

Deploying xApps and rApps exposes the network to software-based threats, including malicious code, AI model poisoning, and data breaches.

Specific Vulnerabilities and Targets

Identified Threats

Security analysis has quantified the threat, identifying a total of 84 distinct threats against the architecture. These range from standard network intrusions, and Man-in-the-Middle attacks, to new, critical risks.

Critical Assets

The security analysis explicitly catalogs the targets that need protection, which are referred to as Critical Assets. A total of 48 assets are identified, broken down into 32 interface assets and 16 logical component assets

Example Threats

spoofed PTP masters causing sync loss, poisoned training data harming slicing, and weak OAuth allowing rogue rApps.

Zero-Trust Model

Open RAN follows a zero-trust approach, meaning every interface must use strong authentication, certificates, and encrypted channels like TLS, SSH, or IPSec.

PART 08

Research Platforms

OpenRAN Gym & Colosseum

OpenRAN Gym

OpenRAN Gym packages the OSC RIC, E2 interface, and an srsRAN RAN into LXC containers, giving an easy platform to test xApps, run experiments, and collect data.

Colosseum Testbed

Colosseum is a large-scale wireless testbed with 128 SDRs and a channel emulator, enabling researchers to conduct large-scale experiments and collect data for ML training.

Integration

The combination of OpenRAN Gym and Colosseum allows for seamless deployment and testing of Open RAN solutions, providing a powerful tool for academic and industrial research.

Arena & PAWR Testbeds

1

Arena Testbed

Arena provides a 64-antenna ceiling grid for indoor 5G testing, enabling research in MEC, private networks, and other 5G use cases.

2

PAWR Platforms

PAWR platforms like POWDER, COSMOS, and AERPAW offer city-scale testbeds for sub-6 GHz, mmWave, and aerial 5G, allowing real-world Open RAN testing.

Open-Source Software Stack

1

srsRAN and OpenAirInterface

Implement 3GPP stacks on USRPs for open RAN deployment and testing.

2

OSC RIC and ONF SD-RAN

Provide near-RT controllers for closed-loop control and optimization.

3

ONAP and OSM

Offer SMO frameworks for orchestration and management of Open RAN components.

4

Full Ecosystem

Together, these tools enable end-to-end Open RAN research, development, and deployment.

References

1. M. Polese, L. Bonati, S. D'Oro, S. Basagni, and T. Melodia, "Understanding O-RAN: Architecture, Interfaces, Algorithms, Security, and Research Challenges," IEEE Communications Surveys & Tutorials, vol. 25, no. 2, pp. 1376 1411, 2023

2. Openess, "05 - Exploring Near-RT RIC and xApps: A Comprehensive Overview," YouTube. Dec. 20, 2023. [Online]. Available: https://www.youtube.com/watch?v=vKurSgb7_tA

THANK YOU!